Watching Me Watching You: How Observational Learning Affects Self-disclosure on Social Network Sites?

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Many explanations have been proposed regarding people's willingness to disclose information on social network sites (SNSs). Focussing on the reciprocal nature of such sites, this study explores the significant role observational learning (OL) plays in determining users' willingness to self-disclose information on Facebook. It demonstrates how the ability to view other users' actions—and the rewards and setbacks they encounter—impinge on their risk assessment and resulting disclosure behavior. Using an online survey of 742 Facebook users and an experiment conducted with 264 such participants, we demonstrated that users learn from others regarding self-disclosure behavior and resulting gains/losses. We showed that the observation mechanism contributes to reward envy, that leads to a high level of self-disclosure behavior. By contrast, observation of risks has only a marginal effect on such undertakings.

Keywords: Social Network Site (SNS), Facebook, Self-disclosure, Observational Learning (OL), Privacy.

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Introduction

Every 60 seconds, 1.09 billion active Facebook users around the world post 510 comments, 293,000 statuses and 136,000 photos. On Twitter, 500 million tweets are submitted each day and billions of comments are shared on other social network sites (SNSs) (Zephoria digital marketing, 2017). This user-created information, that contains personal details, news, moods, and opinions, is critical to sustaining the

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popularity and value of SNSs. Indeed, without massive production and consumption of personal information, SNSs would not be able to maintain the attention required to secure user loyalty (Chen, 2013).

Several approaches have been offered to explain users' motivation to disclose personal information on SNSs. Anticipation of benefits, such as enjoyment and social acceptance, have been suggested as incentives (e.g., Choi & Bazarova, 2015; Sledgianowski & Kulviwat, 2008), whereas perceived possibilities of loss, such as harassment, social rejection, browsing history tracking, third party use of personal data and identity theft, serve as deterrents (e.g., Acquisti, Brandimarte, & Loewenstein, 2015). Our overall motivation is improvement of our understanding of users' willingness to disclose sensitive personal information on SNSs despite the substantial risks that stem from such behavior. More specifically, we aim to advance the current discussion on privacy concerns by tracing the early stages of decision-making processes and exploring how perceptions of gains and losses are learned and acquired. To this end, we point out the significant role the reciprocal nature of SNSs plays in disclosure behavior. We argue that the ability to constantly monitor others and be monitored by others fosters social learning. Drawing on the seminal *observational learning* (OL) *theory* (e.g., Bandura, 1971b; Deutsch & Gerard, 1955; Simpson, Siguaw, & Cadogan, 2008), it is suggested that the visibility of other users' self-disclosure behavior and the gains and setbacks they encounter in the process impinge on users' risk assessment and by implication on their disclosure behavior.

Theoretical framework and hypotheses: self-disclosure behavior in face-to face interactions and on SNSs

Self-disclosure refers to the "process of making the self known to others" (Jourard & Lasakow, 1958, p. 91). SNSs offer many means of fostering and enhancing self-disclosure among users. While almost any action taken on an SNS (e.g., updating profile information, posting a status, photo, or video, clicking on commercial ads) directly or indirectly discloses information about the self (Tsay-Vogel, Shanahan, & Signorielli, 2016), in this study we relate to information SNS users intentionally reveal about themselves.

Studies on self-disclosure—pre- and post-SNS—showed that such behavior encompasses various purposes, which dependent on the setting in which disclosure occurs (e.g., Antaki, Barnes, & Leudar, 2005; Jourard, 1971). Significantly, it has been noted that the decision to share personal information involves counterbalance between two conflicting elements: Anticipated gains and possibility of losses (e.g., Jourard, 1971). It may fulfill fundamental needs, such as social connectedness and belonging (e.g., Tamir & Mitchell, 2012), identity formation and enhancement of self-esteem (e.g., Hollenbaugh & Ferris, 2014), but it also bears potential losses: emotional distresses like jealousy in others' supposedly "good life" life (Muise, Christofides, & Desmarais, 2009), and envy in the rewards they receive online (e.g., rewards like getting support in the form of receiving "Likes"), and other losses associated with undesired usage of the information disclosed that the discloser cannot fully control (Altman, 1975).

To better understand disclosure decisions under conditions of risk and uncertainty, early information privacy scholars coined the term *privacy calculus* (Laufer & Wolfe, 1977). This metaphoric construct constitutes a cost-benefit tradeoff analysis that accounts for the inhibitors and drivers that effect decisions regarding information disclosure. This construct was adopted by scholars who explored disclosure decisions in online interactions, primarily in ecommerce contexts (e.g., Dinev & Hart, 2004; Li, Sarathy, & Xu, 2010). More recently, it was used to explain users' self-disclosure behavior on SNSs (chiefly Facebook). Here, scholars showed that perception of risk was negatively associated with self-disclosure on SNSs, while anticipated benefits were positively associated with it (e.g., Krasnova, Koroleva, Spiekermann, & Hildebrand, 2010; Dienlin & Metzger, 2016; see also McKnight, Lankton, & Tripp, 2011).

While acknowledging the contribution of such works, we identify some weaknesses with regards to their conceptualization of cost-benefit analysis. Specifically, we highlight common tendency in the

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literature to ask users to assess their general perceptions of SNS safety and possible gains (e.g., "overall, I see no real threat to my privacy due to my presence in an SNS" [Krasnova, Veltri, & Günther, 2012]; "Disclosing can improve my living and working efficiency" [Xu, Michael, & Chen, 2013]). Extensive research in risk assessment, however, indicates that people's decision-making processes are far more complex (e.g., Sitkin & Pablo, 1992; Yates & Stone, 1992). When assessing both gains and losses, people consider two dimensions—the significance of a given loss/gain and its probability. For example, winning the lottery could result in a significant profit, but the low likelihood of winning deters people from participating:

> Uncertain consequences or outcomes that might follow one or more options should be evaluated and weighted in reaching an overall decision. Basically, the probability of each outcome is multiplied by the evaluation of that outcome and the score based on these products represents the overall utility of the option. This theory thus provides a model that serves as a rational or a normative basis for making decisions. (Van der Pligt, 2002, p. 258)

This useful perspective was not incorporated in current studies on online self-disclosure. In recent studies, for example, Chang and Chen (2014) and Chang and Heo (2014) offered elaborate scales that measured perceived benefits by using items such as "I feel that Facebook helps me interact with friends." Nevertheless, the authors did not measure perceived importance of interactions with friends via Facebook. Instead, it appears that the study adopts a normative view according to which social ties are universally important. Overlooking the differences among users with regard to significance of specific gains (and losses), results in imprecise estimates of each user's individual privacy calculus. The same shortcoming appears with regard to differences among scenarios. For example, although being hacked is clearly a worrisome situation, it is far less likely to occur than is receipt of comments from friends. By concentrating on magnitude alone, we risk misassessment of the perceived losses users ascribe to these two scenarios.

Having identified these weaknesses, we offer a new conceptualization of cost-benefit analysis in the context of self-disclosure (SD) on SNSs that includes both the likelihood and the magnitude of benefits and costs. We thus hypothesize:

> **Hypothesis 1:** Assessment of gains, the product of user-perceived magnitude and likelihood, will be associated with SD behavior: The more users expect gains by SD, the more they engage in self-disclosure behavior.
>
> **Hypothesis 2:** Assessment of losses, that is also the product of user-perceived magnitude and likelihood, will be associated with SD behavior: The more users anticipate losses by SD, the less they engage in self-disclosure behavior.

The effects of social observation on technology perception and use

As indicated, *OL theory* informs our predictions. In the 1960s and 1970s, Albert Bandura and his colleagues demonstrated that social interactions constitute a significant component in learning processes that can accrue either intentionally or unintentionally. They showed that by observing the behavior of others and the benefits and losses they incur as a result of such actions, one forms an idea of the advisability of a particular behavior. Subsequently, this coded information serves as a guide for action (e.g., Bandura, 1971b). Learning accrues in cases in which the observed model is either actually present or only symbolically so (in mass media artifacts, for example). Significantly, media consumers can learn by viewing a specific role model or from their encounter with a "vast amount of information about human values, styles of thinking and behavior patterns (…) gained from the extensive modeling in the symbolic environment of the mass media" (Bandura, 2001, p. 271). Thus, in the context of mass

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communication, consumers may also learn by watching behavior performed by numerous models, rather than by relying only on one specific model (e.g., Stefanone, Lackaff, & Rosen, 2010).

OL scholarship highlights four dimensions that foster observation learning: *Attention, retention, reproduction*, and *motivation* (e.g., Bandura, 2009). We contend that this four-dimensional framework can be applied successfully to the online environment and especially to SNSs. We begin with the dimension of *attention*, positing that a particular behavior may be learned successfully only if one pays sufficient attention to relevant information regarding, norms, values, thinking patents, conducts, etc., gained from extensive modeling in the media's environment (e.g., Ball-Rokeach & DeFleur, 1976; Bandura, 2001). It is well documented that the attention-drawing mechanism is a major component of popular SNSs (e.g., Hodas & Lerman, 2012). One outstanding example is SNSs algorithms, that attract users' attention to others who share their values and attitudes, while making dissimilar users invisible (e.g., Ellison, Vitak, Gray, & Lampe, 2014; Pariser, 2011). Such a mechanism might reinforce learning, as OL scholars note repeatedly: When models are similar to the learner, the chances that the learner will be influenced by the model's behavior increase (van de Bongardt, Reitz, Sandfort, & Deković, 2015). Another significant "attracting attention" feature is the first-in-first-out queue structure of the platforms' interfaces (e.g., "Time line" in Facebook and chorological queue in Twitter). The second dimension, *retention*, indicates that storing information in ways enabling subsequent access is crucial to the OL process. SNSs might promote retention because these sites constitute archives of social information (Margetts, John, Hale, & Yasseri, 2015) that can be retrieved at different points in time. For example, the individual user can easily utilize the research engine tool placed on top of a Facebook page. By typing a keyword in the designated "research window," they can easily obtain requested information about network members' past behaviors on the network, thereby learning their respective values and attendant benefits and costs. In *reproduction*, the learner gets to practice the observed behavior. SNSs are designed to enhance repetitive SD actions, because their sustainability depends on user disclosure of personal information. Several features enhance such activities. On illustrative example is the ability to "comment on a comment" on one's Facebook page, and clearly observe the chaining patterns. In Tweeter, it is the retweeting feature that facilitate and even encourage reproduction. With regard to *motivation*, observed rewards (gains) and punishment (losses) either raise or lower the learner's motivation to repeat the modeled behavior, as Bandura (1971b) argues: "Seeing others gain desired outcomes by their actions can create outcome expectancies that function as positive incentives [referred to as *perceived gains* in this article]; observed punishing outcomes can create negative outcome expectancies that function as disincentives [referred to here as *perceived losses*]" (p. 276). SNS platforms allow users to "visually monitor" (Carroll & Bandura, 1982) their friends' behavior and the outcomes of these actions. For example, a prime feature in Facebook is the "Like" function. Any Facebook users can easily view the number of "Likes" other people received for the posts they publish. In line with OL scholarship, SNS users will most likely be conform to the behavioral patterns they observe because they expect certain benefits or losses, such as social recognition or rejection, or an increase or decrease in social status.

As SNS may constitute a conducive environment for OL, we aim to explore how the ability to observe other users on SNSs enables one to learn the advisability of the modeled behavior. We develop our hypotheses according to the three core elements users observe and from which they learn: Others' behavior, and the gains—or alternatively, losses—they incur.

Others' SD behavior

Following early theorization of OL, numerous studies documented the influence of observed behavior, stressing the significance of observing behavior performed by many (Bandura, 1971a). As noted by van de Bongardt et al. (2015):

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> The larger the number of peers who engage in a certain behavior, the more functional and correct the behavior will be perceived to be and the more likely it is that adolescents will engage in the same behavior, based on the reasoning that if others are doing it, it is probably a good or wise thing to do the same. (p. 204)

On the face of it, in SNSs such as Facebook, any user can, in principle, visually monitor the behavior of many network friends. Nevertheless, these platforms also embody a visibility constraint: Users who disclose more become more visible. Conversely, risk-averse users usually disclose less and hence are less visible. This may strengthen the perception of self-disclosure as the most common and therefore most prudent course of action.

The influence of others' perceived behavior on users' decisions has been documented in the context of online environments. For example, it was demonstrated repeatedly that in the domain of ecommerce, consumers observe the purchase actions of previous individuals and tend to imitate them (e.g., Wang & Yu, 2017). Similarly, Burke, Marlow, and Lento (2009), who studied online distribution of news, demonstrated that Facebook users who see their friends contributing news content proceed to share more content of this kind themselves (for similar results see Cheung, Liu & Lee, 2015). Building on these insights, we posit:

Hypothesis 3: Perceived SD of others will be positively associated with one's own SD behavior.

Others' gains

As indicated above, one significant element that impinges on the effectiveness of OL is the reward the observed model receives (Bandura, 1971a). In OL theory, if learners see that a model gains from a particular behavior, they develop the expectation that such behavior would afford them the same gains. In such instances, users might be motivated to perform the observed behavior. Although observed gain is described as a fundamental factor in OL, to the best of our knowledge, it has been overlooked in SD literature on SNSs. Recent studies acknowledged the importance of observing behavior of others, as well as the gain one receives directly from the learning process (Cheung et al., 2015), yet no research has addressed the link between the two components. Hence we propose the following:

Hypothesis 4: Observed gains of others will be positively associated with one's perception of gains to self.

With regard to *others' losses*, OL theory posits that punishments (or losses) incurred by others receive deter the learner from repeating the observed behavior (van de Bongardt et al., 2015). Recently, this idea was tested in a study on software piracy, in which scholars showed that people tend

Figure 1 Suggested theoretical model.

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to avoid software pirating when they see online that those who commit this offence suffer losses (e.g., Sun, Lim, & Fang, 2013). In accordance with these observations, we propose (Figure 1):

Hypothesis 5: Observed losses of others will be positively associated with one's assessment of losses.

Study 1: survey design

Method

We conducted an online survey among Israeli Facebook users between 7 and 9 December 2015.

Sample

We used the service of an online survey company that holds an online panel of participants who receive gift vouchers in exchange for their participation in online surveys. For this study, a sample of 742 Israeli Facebook users was created. The sample is not representative of the population of Israel, but rather was designed to be representative of the Israeli population of Facebook. Marketing data[1] show that this latter group is far younger than the population as a whole and comprises more men (53%) than women. Israeli Facebook users are also more secular than Israeli society as a whole.[2] The sample constitutes a near-perfect match to data available with regards to age and gender composition of Israeli Facebook users (age ranging between 18 and 71 with $M = 42.3$ $SD = 14.6$; 54% male; 3% ultra-Orthodox; 45% holding an academic degree; income: $M = 3.0$ $SD = 1.2$, on a 5-point scale).

Measures

For a full description of scale items and validation, see Appendix A.

Dependent variable[3]

To measure SD behavior, we created two scales: *Actions taken* and *content shared*. For the first scale, we asked participants to rank how often they were involved in a list of activities (adapted from Tsay-Vogel et al., 2016; 9 items, Cronbach's alpha = .88), while the second inquired how often participants shared content concerning different aspects of their life (e.g., politics, consumption preferences, family, etc.; Cronbach's alpha = .88). In both cases, answers were expressed on a scale ranging from 1 (never) to 5 (more than once a day) (actions: $M = 2.4$, $SD = .86$; information sharing: $M = 1.7$, $SD = .70$).

Independent variables

Perceived SD of others. To measure participants' perception of how much others disclose, we asked them to answer the same questions as above with regard to their Facebook friends (Actions: Cronbach's alpha = .90; $M = 4.0$, $SD = .94$; information sharing: Cronbach's alpha = .93; $M = 3.1$, $SD = 1.0$).

We measured significance and likelihood of both gains and losses (Yates & Stone, 1992).

Perception of gains. To gauge *perception of gains* that could be attributed to SD, we created a list of possible benefits (Xu et al., 2013). To assess their significance, we asked participants the following question: "To what extent do you consider the following to be advantages of Facebook use?" Answers were ranked on a scale from 1 (not an advantage at all) to 5 (a significant advantage) (6 items, Cronbach's alpha = .84; $M = 3.3$, $SD = 1.0$). To assess likelihood, we asked: "When you think about the list we mentioned earlier, to what extent do you believe you are able to make the most of these benefits?" Answers were ranked on a 1 to 5 scale, ranging between "not at all" to "very much" (6 items, Cronbach's alpha = .87; $M = 3.2$, $SD = .96$).

Perception of losses. To measure losses possibly attributable to SD, we created a list of possible types of harm (Acquisti et al., 2015). As in the case of gains, we asked participants about magnitude, i.e., the extent to which they considered the listed events to be troubling: The answer scale ranged from 1 (not troubling at all) to 5 (very troubling) (7 items, Cronbach's alpha = .87; $M = 3.8$, $SD = .97$). We also asked participants to evaluate the likelihood of each loss: "How likely is it that

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such things will happen to you as a result of your Facebook use?" Answers were ranked on a 1 to 5 scale, ranging between "not at all" to "very" (7 items, Cronbach's alpha = .91; $M = 3.6$, $SD = .99$).

Observed losses and gains of others. To measure participants' assessments of the gains others achieved by using Facebook, we asked: "In your opinion, how successful are your Facebook friends in making the most of these benefits?" using the same list as above. The response scale ranged from 1 (not at all successful) to 5 (very successful) (6 items, Cronbach's alpha = .90; $M = 4.0$, $SD = .83$).

To gauge participants' evaluation of others' losses, we also asked whether any of the above risks affected their Facebook friends. Responses reflected a 3-point scale: "Never," "once" and "more than once" (ranging from 0 to 7, $M = 3.3$, $SD = 2.8$).

Covariates

Besides applying demographic criteria, the model used three questions to control for online behavior. Participants were asked: (a) if, and how often they browse the web, (b) if, and how often they access Facebook, and (c) if, and how often they access other SNSs, such as Instagram or Twitter. Answers ranged from 1 (never) to 5 (several times a day). Participants who reported they engage in any of these activities on a daily basis were then asked how many hours a day they spent doing so, with answers reflecting a 3-point scale: "Less than an hour," "1–3 hours," and "more than 3 hours." In our sample, only 1% reported browsing and 11% reported using Facebook on a less than daily basis. Consequently, we only used the hours per day item to report web browsing behavior (for which the 1% that did not answer this question were coded as 0, $M = 2.6$, $SD = .60$). For Facebook and other SNS use, we created a combined 6-point scale based on the two measures (Facebook $M = 4.8$, $SD = 1.0$; other SNS $M = 2.7$, $SD = 1.0$).

Past experience

We also asked participants whether any of the above risks affected them, with responses ranging on a 3-point scale: "never," "once," and "more than once" (ranging from 0 to 7, self: $M = 3.1$, $SD = 2.0$).

Statistical analysis

As indicated above, this theoretical approach to cost-benefit analysis calls for a process of weighting the significance of each item in gain/loss scales according to its perceived likelihood ([Van der Pligt, 2002]; gains: $M = 11.8$, $SD = 5.5$; losses: $M = 14.1$, $SD = 5.9$, see Appendix B for distribution). The theoretical model described above requires mediation relations testing, for which we calculated Maximum Likelihood (ML) estimates, using IBM SPSS Amos 22.0 to assess the proposed model's goodness of fit and test each of the hypotheses. To test mediation, we relied on bootstrapping (5,000 bootstrap samples) and following MacKinnon, Lockwood, Hoffman, West, and Sheets (2002) we opted for a bias-corrected confidence interval (Preacher & Hayes, 2008).

Results

Before testing our hypotheses, we note the correlations between the two dimensions mentioned—significance and likelihood of gains and losses—and SD behavior. In the case of gains, a strong positive correlation was found between significance and likelihood of gains and SD (significance: acts: Pearson's $r = .49$, $P < .01$, information: Pearson's $r = .35$, $P < .01$; likelihood: acts: Pearson's $r = .55$, $P < .01$, information: Pearson's $r = .41$, $P < .01$). By contrast, in the case of losses, both diminutions were weakly correlated with SD behavior (significance: acts: Pearson's $r = .02$, $P = .75$, information: Pearson's $r = -.07$, $P = .07$). Importantly, likelihood was positively correlated with it (acts: Pearson's $r = .07$, $P = .08$, information: Pearson's $r = .08$, $P = .07$).

We used Structural equation modeling (SEM) to test our hypotheses. The model was assessed with SD actions and sharing information creating a latent variable of SD behavior for both self and others (for full correlations table, see Appendix C). The theoretical model yielded satisfactory results

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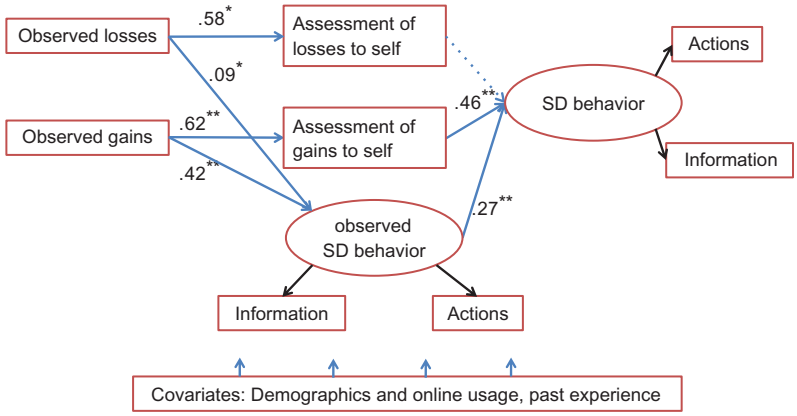

Figure 2 SEM of SD behavior social learning.
Notes: Parameters are standardized coefficients. $^{***}p < .001$, $^{**}p < .01$, $^{*}p < .05$; N = 742.

in terms of goodness of fit indices. We obtained a chi-square to df ratio (CMIN/DF) of 1.58 and found that the model fit the data extremely well ($\chi^2 = 55.4$, $df = 35$, $p = .02$; RMSEA = .03, CFI = .987).

Figure 2 shows that most of our hypotheses were confirmed. Insofar as gains are concerned, H1 and H4 were supported: observed gains were positively associated with gains assessment for self (H4), that were, in turn, positively associated with SD behavior. Observed gains had a positive and significant indirect effect on SD behavior (standardized estimates = .39; 95% C.I. [.21–.32]).

As for losses, H5 was supported, with observed losses to others positively associated with assessment of loss to self. Contrary to H2, assessment of losses to self had no bearing on SD behavior. Finally, in keeping with H3, perceived SD of others behavior was positively associated with the participants' own activity in this respect.

It is important to note that results of Study 1 are the product of a cross-sectional design. As such, they are open to alternative interpretations and do not establish a causal relation.

Study 2: experimental design

The primary advantage of our survey design is the ability to gather information on users' perceptions of their social reality. The survey, however, cannot establish causality, but can only demonstrate the association among the variables. In the case at hand, there could be two alternative explanations to the observed correlations between perceptions of others' behavior gains and losses and cost-benefit assessment for self. One option is that users learn from others (wherein the direction of causal relations is from others to self). Alternatively, it could be interpreted as a projection process in which one projects one's own views on others. Consequently, we conducted an online experiment to demonstrate that OL does occur. We exposed participants to one of the most common gain and loss scenarios that Facebook users encounter: The sentiments expressed in responses to a given post (positive vs. negative).

Method

Participants

Using the same recruitment platform, we sampled 512 participants for the experiment. Of these, 44 did not have a Facebook account and were therefore excluded. An additional 41 did not complete the

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Figure 3 Experimental stimuli.

experiment, leaving 427 participants in the final sample (age ranged between 18 and 71 with $M = 37$ $SD = 13.1$; 59% female; 2% ultra-Orthodox; 42% with academic degrees; income: $M = 2.8$ $SD = 1.2$ on a 5-point scale).

Manipulation

Participants were divided randomly into four groups. Each was exposed to a segment of a Facebook page (see Figure 3) presenting a news feed in which a person shared a post and the reactions it received. In all cases, the post was identical, but the responses to it were manipulated. In the control case, there were no comments and 15 *Likes*.[4] In the next three cases, we manipulated the type of responses, but held the amount of attention constant (that is, the number of comments and Likes were the same). In all cases, two comments were identical, one positive and one negative, while four additional comments varied in sentiment. The comments addressed four issues: Characteristics, post content, political issue described and public support regarding the issue (whether or not the post represents the opinion of many or few). The picture, user names of people commenting and length of comments were held constant across all cases. In the positive cases, the post received Likes and positive emoticons, as well as supportive comments from other users. In the negative case, responses included negative emotions and comments. Finally, a mixed response case included both positive and negative emotions, along with three positive and three negative comments.

Manipulation check. To ensure that participants noticed the manipulation, we included five statements at the end of the questionnaire and asked participants to indicate the extent of their agreement on a scale of 1 (disagree) to 5 (agree): (a) The comments to the post were unpleasant, (b) The comments to the post were supportive (excluding participants in the control, $F_{(2,425)} = 13.9$, $p < .01$; $F_{(2,425)} = 32.0$, $p < .01$), (c) It was important to post this content ($F_{(3,423)} = 5.5$, $p < .01$), (d) The post was convincing ($F_{(3,423)} = 5.1$, $p < .01$), and (e) The post was inappropriate ($F_{(3,423)} = 1.1$, $p = .40$). In all but the last case, the differences among groups were significant and conformed to the anticipated directions (for means and std. errors, see Appendix D), suggesting that the manipulation worked.

Measures. After exposure, participants were instructed to respond according to the scales described above for perceived losses (Cronbach's alpha = .86; $M = 3.8$, $SD = .90$) and gains (Cronbach's alpha = .86; $M = 3.4$, $SD = .94$) and were asked how likely they were to self-disclose, using the same items as earlier (Cronbach's alpha = .86; $M = 2.6$, $SD = .83$). The order of scales was randomized across participants and ANOVA was conducted to test the hypothesis.

Results

Figure 4 presents the means and standard errors of all dependent variables, according to experimental case. As anticipated, participants in the positive case group reported a higher tendency to share when

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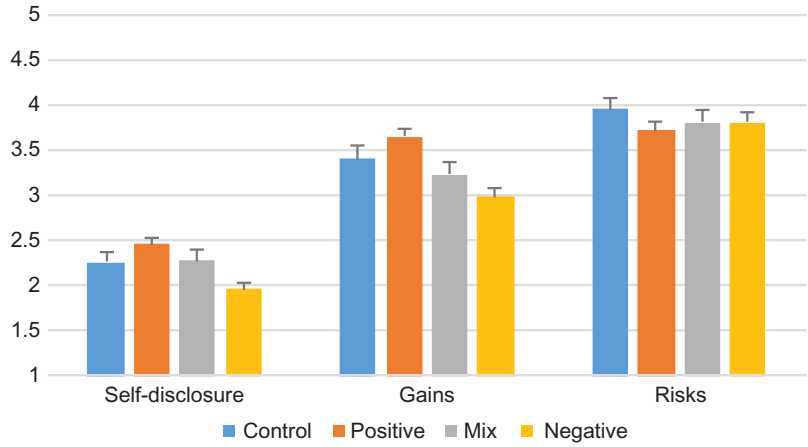

Figure 4 Means and standard errors: risk perception, gains perception and tendency towards SD behavior, by experimental case. Scale ranges from 1 (not at all) to 5 (a lot).

using SNSs, while those in the negative case group reported a lower one ($F_{(3,424)} = 5.5$, $p < .01$; all differences among groups are significant; for a full description of contrasts, see Appendix E). Perceptions of gains were also affected by the manipulation in the expected direction ($F_{(3,424)} = 7.37$, $p < .01$). Participants in the positive case group perceived a greater number of advantages to self-disclosure behavior (in comparison to the negative case and the mixed case) while those in the negative case group reported the opposite (in comparison to the control group and positive case group).

By contrast, there was no significant difference in perception of losses ($F_{(3,424)} = .70$, $p = .55$); contrary to expectations, participants in the negative case group did not demonstrate a higher perception of possible losses. We also observed correlations among the three dependent variables. In the current study, the correlation between SD tendency and perceived gains remains much stronger than the one between SD tendency and perceived losses (gains: $r = .53$, $p < .001$; losses: $r = -.08$, $p = .20$).

We conjectured that OL shapes Facebook users' perceptions of losses and gains, wherein users learn how much to share and the outcomes of various types of sharing behavior by observing others online. Most of these hypotheses were largely supported by empirical data: Surveys and experiments alike demonstrated that Facebook users learn from others regarding gains/losses and behavior. Moreover, perceived gains were positively correlated with SD behavior. Only partial confirmation was obtained concerning losses, however: While a significant association was found between perceived losses to self and to others, the applied manipulation did not alter perception and no effects on behavior were noted.

Discussion

Our study applied the privacy calculus perspective to assess the influence of OL on SD in SNS. We conceptualized a model that underscores three key mechanisms at work: (a) Facebook users' association of losses and gains with SD on SNS, (b) their assumption that positive relations obtain in both cases, and (c) the powerful effect of OL on losses/gains perception and—by implication—on SD behavior. By focussing on OL's role in determining SD perceptions and resulting behavior, this model adds a new dimension to existing studies emphasizing the effects of social influence on SD (e.g., Shibchurn & Yan, 2015). A core argument of these studies maintains that significant others are likely

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to contribute to the formation of one's intentions regarding SD behavior. For example, Shibchurn and Yan (2015) recently speculated that "[h]igh risk perceptions and low disclosure levels by significant others may lead individuals to disclose less" (p. 106). If significant others disclose information plentifully, users are likely to adopt such patterns and disclose more. To date, their hypothesis has not been tested empirically. We attempted to advance this discussion by focussing on aspects of social influence that ought to be accorded more attention in an age of networked media: rather than focussing on perceived social norms (what one believes others think), we addressed technological features that enable SNS users to retrieve information regarding others by viewing their conduct and the consequences thereof. We were thus able to shed new light on the complex effect of social influence of risk assessment on disclosure behavior.[5]

We discovered that when users perceive other Facebook users' self-disclosure, together with the benefits they obtain thereby, they tend to believe that they too can leverage the platform to their own advantage. We showed that this conception—that can be summarized as "more self-disclosure = greater benefit"—encourages participants to intensify their SD behavior. The situation differs with regard to perception of losses. We found that assessment of losses has a marginal effect on users' SD behavior. Although the first study (survey) demonstrated that users' risk assessment was linked with their perceptions of other users' losses, the second (experiment) did not replicate these results. In fact, manipulation influenced assessment of gains and not of losses. These findings are in line with existing scholarship indicating that although users ascribe high risk levels to SD (lack of safety, security, reliability and trustworthiness), they still disclose a considerable amount of apparently sensitive information (e.g., Acquisti & Gross, 2006; Taddicken, 2013). When predicting Facebook self-disclosure, contemporary scholars studying privacy calculus on SNSs noted that expected benefits have more predictive power than perceived losses (Dienlin & Metzger, 2016).

We suggest two reasons that might explain the difference between the predictive power of gains and losses on SD. First, as noted by Ellison et al. (2014) in their study of Facebook, this platform is designed to impel users to reward those who generate content, because the absence of response is a vague signal that might be interpreted as a lack of interest. In Ellison et al.'s words:

> SNS users must respond in a manner that leaves an observable marker of attention as a way of cultivating, or grooming, their connections. (…) On an SNS, explicitly responding to another user via activities that leave visible traces, such as commenting or clicking the "Like" button, is the most reliable way to indicate one has seen and attended to any individual piece of content on the site. (p. 858)

Following this reasoning, gains may well become more apparent in users' feeds, leading them to believe that other members of their network leverage the platform more than they do. We argue that this perception of the rewards other users are supposedly receiving is among the factors that motivate Facebook users to disclose information liberally.

Another possible reason for the observed disparity concerns our approach to studying risk assessment in connection with SD on Facebook. Our findings suggest that in the case of perceived gains, the significance and likelihood of the gain have a similar effect on SD, but when it comes to losses, the picture is more complex: The significance of a given loss is negatively correlated with SD, whereas the likelihood of a given loss is positively correlated with it. This may well reflect the complexity of risk-assessment as it applies to SD on Facebook: Those who attribute significant losses to SD on Facebook will be less likely to self-disclose, regardless of their perceived likelihood of these losses. By contrast, those who disclose more may realize that these activities increase their chances of being harmed, but they might not necessarily perceive the possible losses as significant.

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Limitations and future research directions

Several aspects of the research design may well have affected the findings of this study. Measures, for example, may constitute a significant source of bias. To reinforce design, we supplemented the survey with an experiment. Although the resulting combination provides powerful methodical strength, alternative designs could have been considered as well, such as a study of organic interactions on Facebook, analyzing the interplay between visible social information and behavior patterns, made possible by the advent of unobtrusive behavioral measurement. Its advantages notwithstanding, this method also entails certain constraints, particularly ethical concerns: when monitoring SNS personal profiles of consenting participants, for example, exposure of research to content posted by others who provided no such consent is virtually inevitable. Furthermore, this study used an opt-in sampling method, wherein participants selected themselves. Obviously, a non-random sample of paid participants would engender various biases. Considering the extremely low response rates prevailing at present, however, the method appeared particularly useful despite its limitations.

Another constraint worth mentioning involved the characteristics of the scales used. This study comprises some scales that contain "and" and "or" clauses. For example, we asked participants if they share personal information on Facebook, but did not specify whether this information contains a phone number, an email address, a password, etc. Clearly, while users might be willing to share phone numbers, they would not necessarily agree to sharing passwords. Future studies could focus on specific SD behavior and nuance our scales. Related to this, is a possible discrepancy between the use of the word "importance" of given benefit users report to, and the "gain" they attribute to it. For example, getting "Likes" might be a big benefit of SD on SNSs, but a respondent might not see it as "important" since the word might be associated with issues that carry social importance. Future research can address this limitation by "fine-tuning" the measurement.

Several additional factors may have exerted a substantive effect on results. For example, the sample was limited in two respects, as it comprised Israeli users alone and was tailored to one platform only—Facebook. Although Facebook is the most popular SNS in Israel (Alexa, 2017), the study should be replicated in other cultural/national settings in which users may have a different predisposition regarding self-disclosure. Such replication could also be useful in countries whose privacy regulations and concerns differ from those of Israel. Considering the well-documented dissimilarities among countries (see, e.g., Krasnova & Veltri, 2010; Ribak & Turow, 2003), one would expect users in countries with different approaches to privacy to ascribe differential gain/losses levels to SD on SNS. Studies of this kind might provide more intensive insight into the effects of culture on cost-benefit analysis and resulting SD behavior. Similarly, future efforts could consider the manner in which designated visibility features inherent in certain online platforms such as *WhatsApp, Instagram*, or *LinkedIn* affect users' perceptions of others and their behavior patterns.

Participants' ages may also contribute to research bias. Our sample comprises users of different ages (see discussion in the Methodology section), as we sought to provide a representative (nationwide) sample of Facebook users (to whatever extent possible—demographic information concerning Facebook users is not available to the public). It may be valuable, however, to conduct a comparative study with cohorts representing different age groups (Generation Y and the older Generation X and Baby Boomers). As an increasing number of studies show that privacy concerns among young people differ from the anxieties manifested by adults, subsequent studies could shed considerable light on the manner in which participant age affects the value ascribed to SNS losses and gains. In this study, we did not distinguish between strong tie and weak tie friends. Realizing that social tie strength might influence OL, it could be valuable to conduct a study that explores the extent to which it does so.

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The study might also contribute to practice-led research. One issue that received massive attention by both researchers and practitioners is means of fostering privacy literacy in the context of online networked technologies, particularly on SNSs. The findings of the current study, that underscored the importance of reciprocal features of SNS, could assist in the construction of interventions that enhance users' capability to control the effects of their disclosure behavior. At present, Facebook responds to this challenge by sending its users personal reminders to review their privacy settings. One possible intervention that emerged from this study is an interface design that provides a visual display of friends' privacy-related decisions, such as the relevant privacy settings.

This study contributes to a growing body of research on SD behavior by pointing to the significant role OL plays in determining users' willingness to self-disclose personal information on SNSs. It shows that the observation mechanism contributes to *reward envy*, that leads, in turn, to a high level of online SD. Subsequent studies could apply the model to examine the effects of OL on various manifestations of human conduct in online spaces.

Acknowledgments

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Notes

1 See http://www.fialkov.co.il/english/facebook-audience-insights, accessed 23 December 2017.
2 Ultra-Orthodox Jews, who comprise 8.8% of the population over age 20, often abstain from Internet use for ideological and economic reasons. See CBS, Statistical Abstract of Israel 2013—No. 64, Subject 7, Table 7.1, *Persons aged 20 and over by selected characteristics*., accessed 2 September 2014. Because of language barriers, the Israeli Arab population was not included in the sample.
3 The order of scales was randomized across participants.
4 We chose to present a certain degree of social response, given that typically any post receives some attention, thus a post with no response at all may appear unrealistic.
5 Our findings show that users discern positive relationships between levels of self-disclosure and of gains/losses. When observing others' gains, users appear to perceive self-disclosure engagements on Facebook as a double-edged sword, exacerbating personal vulnerability as it increases platform usefulness.

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Appendix A: Measurements

Confirmatory factor analysis

SD acts—self

act1 Write status updates
act2 Comment on other people's status updates
act3 Share other people's content
act4 Upload photos
act5 Tag yourself in photos
act6 Tag other people in photos
act7 Participate in groups
act8 Indicate your geographical location, for example by mentioning a name of a restaurant, an event, or a landmark
act9 Indicate you intend to go to an event, or check in

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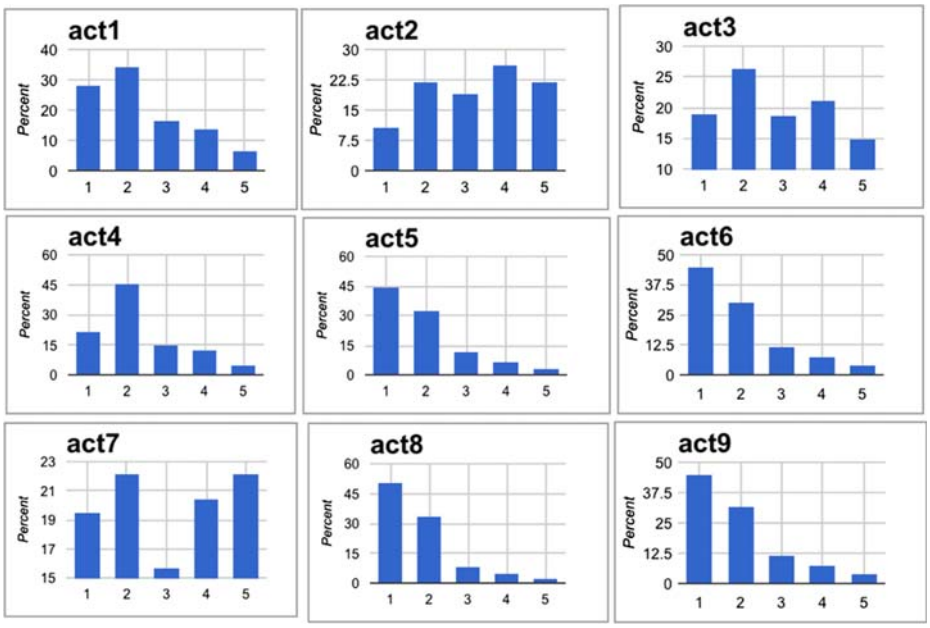
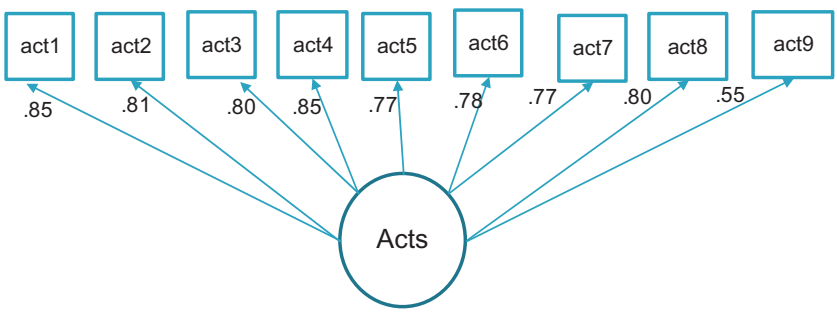
	N	Min.	Max.	Mean	Std. D.	Skewness	Kurtosis
act1	744	1	5	2.36	1.22	.64	−.60
act2	744	1	5	3.26	1.32	−.20	−1.16
act3	744	1	5	2.86	1.35	.15	−1.20
act4	744	1	5	2.32	1.10	.79	−.09
act5	744	1	5	1.91	1.07	1.20	.79
act6	744	1	5	1.96	1.13	1.13	.45
act7	744	1	5	3.04	1.45	−.01	−1.38
act8	744	1	5	1.75	.97	1.46	1.79
act9	744	1	5	1.94	1.11	1.17	.60



Fit of model:

$\chi^2 = 34$, $df = 11$, $p = .46$; RMSEA = .00, CFI = .997



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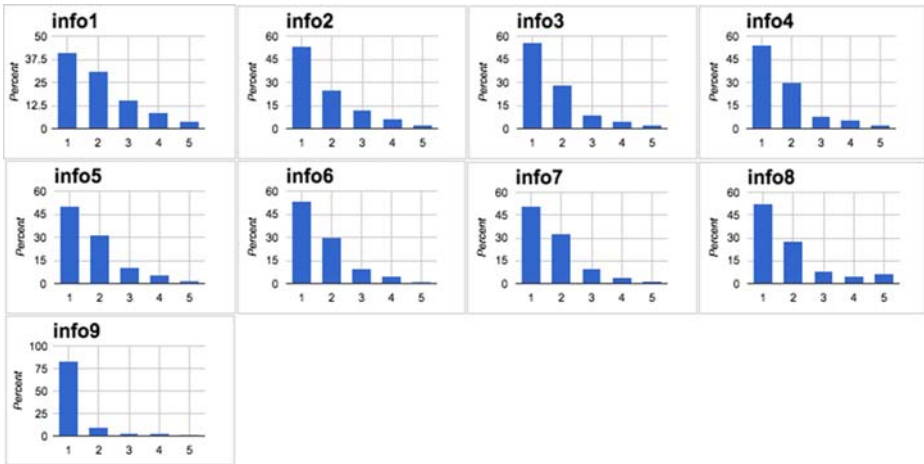
SD information-self

info1 Share content regarding politics and current events, such as your opinions, political content you support, or news stories you found interesting
info2 Share content regarding your consumer preferences and shopping habits, such as companies and products you like or dislike
info3 Share information regarding your family, such as your relationship status, the identity of your partner, your parental status, or activities in which your children took part
info4 Share information regarding your work, such as your workplace, your profession, or the people you work with
info5 Share information regarding your friends, such as the identity of the people you hang out with
info6 Share information regarding your current location, such as restaurants or events you are currently attending
info7 Share information regarding your location in retrospect, such as restaurants or events you previously attended
info8 Indicate where you live on your profile
info9 Share personal information, such as phone numbers, ID numbers, or passwords, or send it in messages

	N	Min.	Max.	Mean	Std. D.	Skewness	Std. Error	Kurtosis	Std. Error
info1	744	1	5	2.04	1.13	.94	.09	.04	.18
info2	744	1	5	1.78	1.04	1.31	.09	.99	.18
info3	744	1	5	1.7	.98	1.54	.09	1.92	.18
info4	744	1	5	1.71	.98	1.53	.09	1.88	.18
info5	744	1	5	1.78	.98	1.31	.09	1.19	.18
info6	744	1	5	1.69	.92	1.37	.09	1.44	.18
info7	744	1	5	1.72	.91	1.39	.09	1.74	.18
info8	743	1	5	1.85	1.16	1.46	.09	1.27	.18
info9	744	1	5	1.3	.78	2.99	.09	8.78	.18



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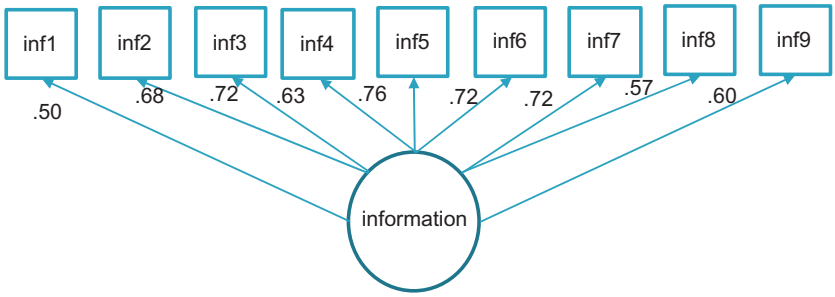
Fit of model:

$\chi^2 = 15.9$, $df = 26$, $p = .87$; RMSEA = .00, CFI = .995



SD acts—Others

act_fr1 Write status updates
act_fr2 Comment on other people's status updates
act_fr3 Share other people's content
act_fr4 Upload photos
act_fr5 Tag themselves in photos
act_fr6 Tag other people in photos
act_fr7 Indicate their geographical location, for example by mentioning a name of a restaurant, an event, or a landmark
act_fr8 Indicate they intend to go to an event, or check in
act_fr9 Participate in groups

	N	Min.	Max.	Mean	Std. D.	Skewness	Std. Error	Kurtosis	Std. Error
act_fr1	698	1	5	3.97	1.04	−.80	.09	−.22	.19
act_fr2	697	1	5	4.21	.97	−1.21	.09	.91	.19
act_fr3	693	1	5	3.98	1.04	−.87	.09	.01	.19
act_fr4	710	1	5	3.94	1.04	−.80	.09	−.15	.18
act_fr5	668	1	5	3.71	1.14	−.56	.10	−.65	.19
act_fr6	668	1	5	3.62	1.16	−.51	.10	−.67	.19
act_fr7	677	1	5	3.46	1.20	−.37	.09	−.84	.19
act_fr8	650	1	5	3.44	1.18	−.34	.10	−.84	.19
act_fr9	633	1	5	3.93	1.07	−.83	.10	−.10	.19

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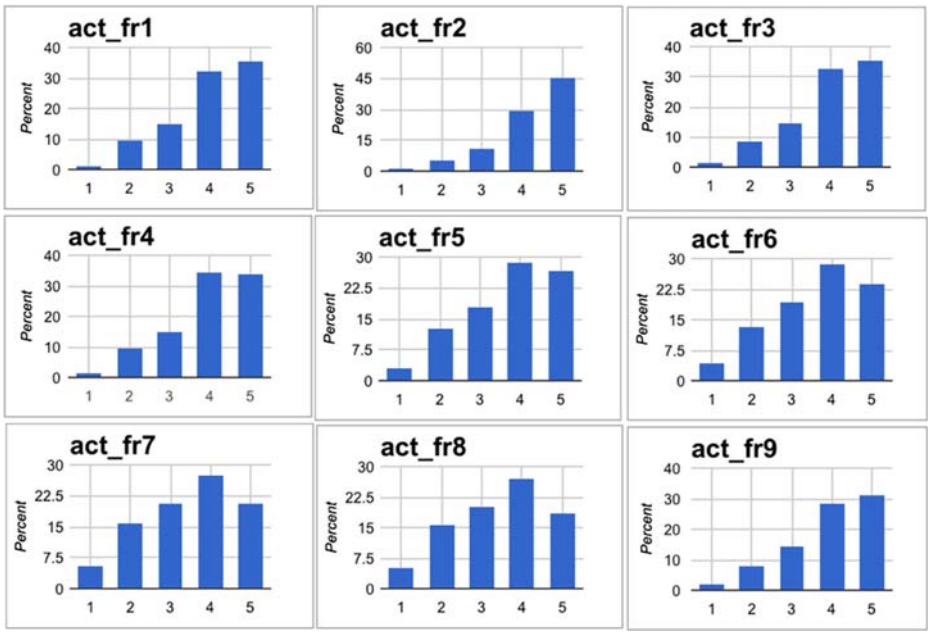
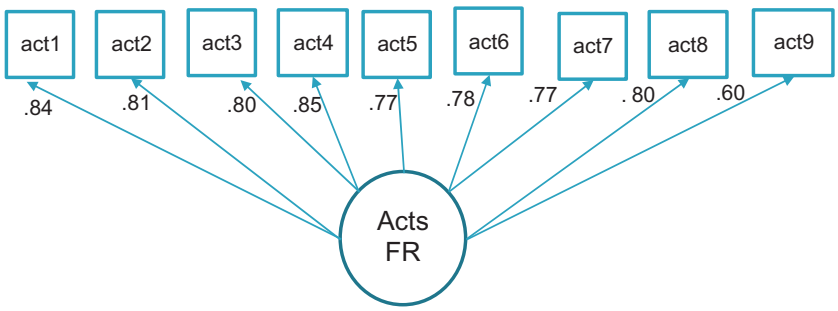


Fit of model:

$\chi^2 = 37.5$, $df = 15$, $p = .07$; RMSEA = .03, CFI = .981



SD-information others

info_fr1 Share their opinions regarding current events and political contents they support

info_fr2 Share content regarding their consumer preferences and shopping habits, such as companies and products they like or dislike

info_fr3 Share information regarding their family, such as their relationship status, the identity of their partner, their parental status, or activities in which their children took part

info_fr4 Share information regarding their work, such as their workplace, their profession, or the people they work with

(*Continued*)

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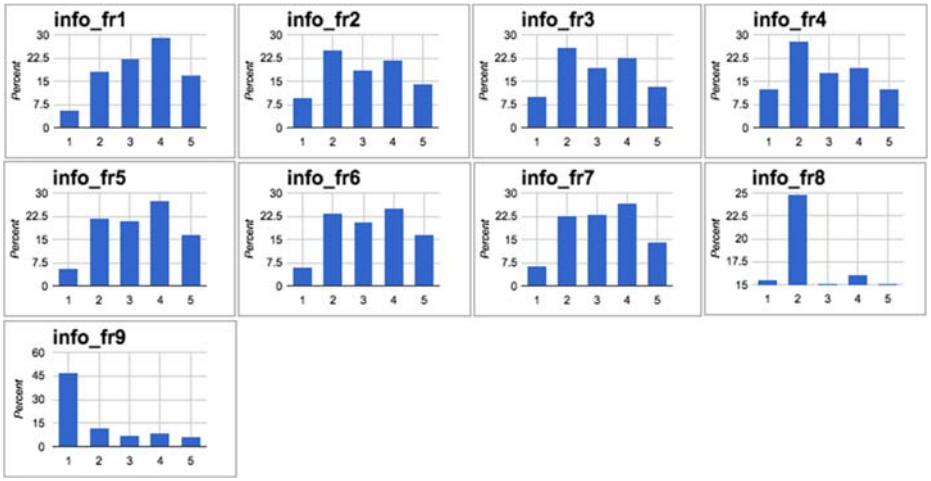
Continued

info_fr5 Share information regarding their friends, such as the identity of the people they hang out with or how they hang out with them
info_fr6 Share information regarding their current location, such as restaurants or events they are currently attending
info_fr7 Share information regarding places they previously attended, such as restaurants and events
info_fr8 Indicate where they live on their profile
info_fr9 Share personal information, such as phone numbers, ID numbers, or passwords

	N	Min.	Max.	Mean	Std. D.	Skewness	Std. Error	Kurtosis	Std. Error
info_fr1	688	1	5	3.37	1.17	−.28	.09	−.86	.19
info_fr2	669	1	5	3.06	1.26	.03	.09	−1.12	.19
info_fr3	681	1	5	3.03	1.25	.05	.09	−1.09	.19
info_fr4	676	1	5	2.91	1.28	.18	.09	−1.11	.19
info_fr5	690	1	5	3.3	1.19	−.16	.09	−1.01	.19
info_fr6	684	1	5	3.25	1.21	−.10	.09	−1.06	.19
info_fr7	692	1	5	3.21	1.17	−.10	.09	−.95	.19
info_fr8	644	1	5	2.89	1.37	.19	.10	−1.23	.19
info_fr9	611	1	5	1.96	1.34	1.12	.10	−.15	.20



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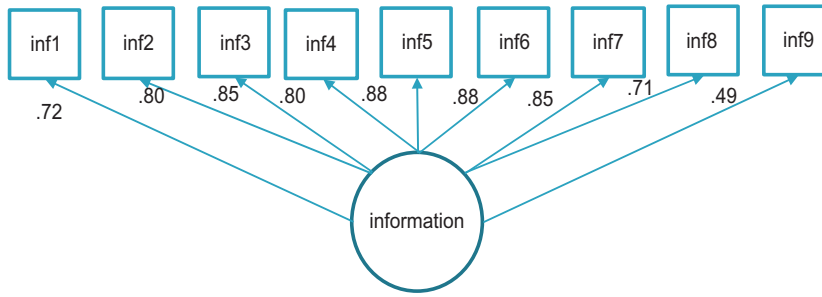
Fit of model:

$\chi^2 = 10.3$, $df = 16$, $p = .85$; RMSEA = .00, CFI = .994

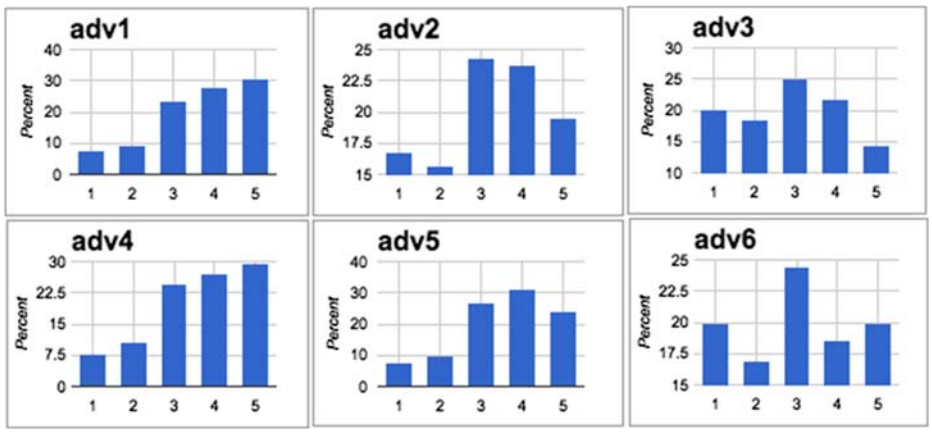


SD-Gains self:

adv1 Simply and rapidly obtaining information suiting my interests
adv2 Obtaining likes and support from others
adv3 Expressing myself—my opinions and feelings
adv4 Maintaining and reinforcing social relations
adv5 Getting personalized information, such as information about my close friends or about my fields of interest
adv6 Effectively promoting myself professionally

Significance

	N	Min.	Max.	Mean	Std. D	Skewness	Std. Error	Kurtosis	Std. Error
adv1	744	1	5	3.64	1.23	−.63	.09	−.52	.18
adv2	744	1	5	3.13	1.35	−.18	.09	−1.13	.18
adv3	744	1	5	2.92	1.33	.01	.09	−1.15	.18
adv4	744	1	5	3.60	1.24	−.56	.09	−.63	.18
adv5	744	1	5	3.53	1.19	−.54	.09	−.48	.18
adv6	744	1	5	3.02	1.40	−.03	.09	−1.23	.18



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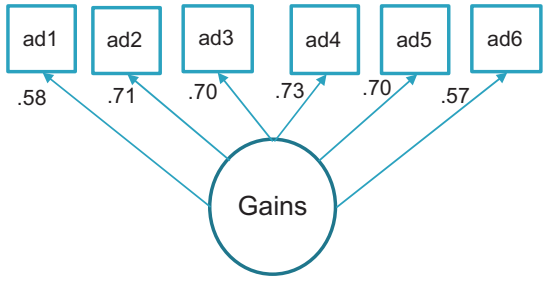
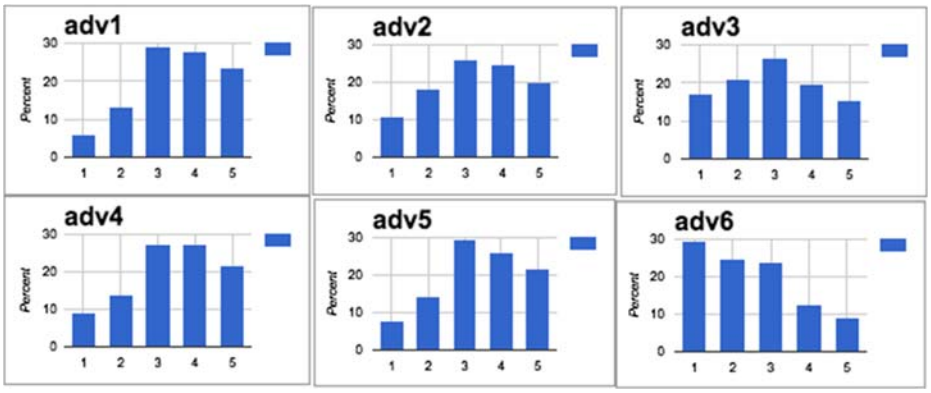
Fit of model:

$\chi^2 = 10.3$, $df = 5$, $p = .08$; RMSEA = .03, CFI = .995



SD-Gains self:
Likelihood

	N	Min.	Max.	Mean	Std. D.	Skewness	Std. Error	Kurtosis	Std. Error
adv1	743	1	5	3.49	1.17	−.37	.09	−.67	.18
adv2	743	1	5	3.24	1.27	−.20	.09	−.99	.18
adv3	743	1	5	2.96	1.31	.04	.09	−1.08	.18
adv4	743	1	5	3.38	1.23	−.36	.09	−.79	.18
adv5	743	1	5	3.40	1.20	−.31	.09	−.75	.18
adv6	743	1	5	2.47	1.29	.48	.09	−.84	.18



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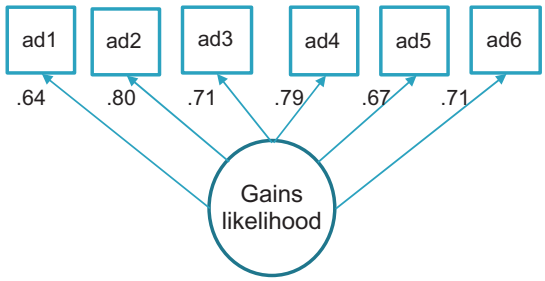
Fit of model:

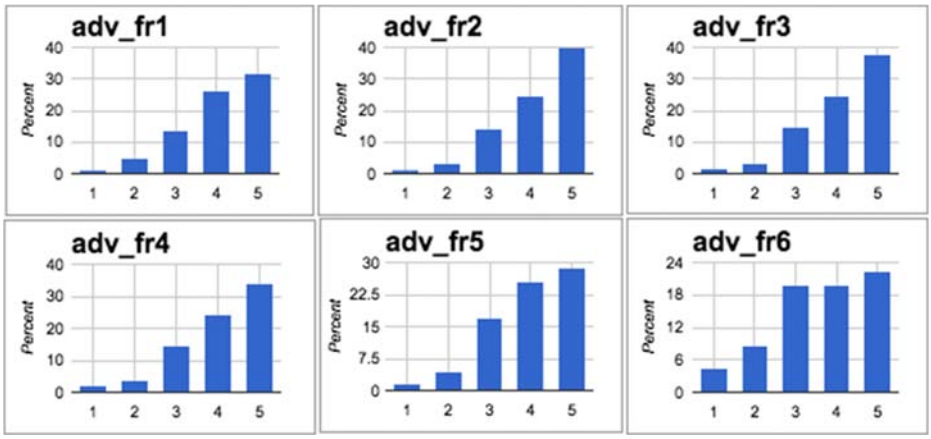
$\chi^2 = 5.39$, $df = 5$, $p = .14$; RMSEA = .033, CFI = .998



SD-Gains others:

adv_fr1 Obtaining information suiting their interests
adv_fr2 Obtaining likes and support from others
adv_fr3 Publicly expressing themselves and their feelings
adv_fr4 Reinforcing social relations
adv_fr5 Getting personalized information, such as information about their close friends or about their fields of interest
adv_fr6 Promoting themselves professionally

	N	Min.	Max.	Mean	Std. D.	Skewness	Std. Error	Kurtosis	Std. Error
adv_fr1	579	1	5	4.07	.98	−.91	.10	.24	.20
adv_fr2	620	1	5	4.18	.95	−1.08	.10	.71	.20
adv_fr3	614	1	5	4.13	.99	−1.05	.10	.62	.20
adv_fr4	589	1	5	4.07	1.02	−1.02	.10	.53	.20
adv_fr5	575	1	5	3.98	1.00	−.77	.10	.04	.20
adv_fr6	560	1	5	3.63	1.19	−.51	.10	−.62	.21



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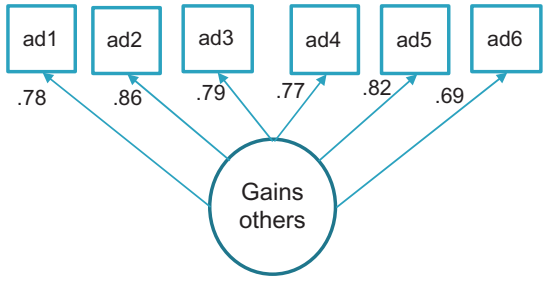
Fit of model:

$\chi^2 = 3.2$, $df = 5$, $p = .68$; RMSEA = .00, CFI = .998



SD loss—self

rsk1 Information I want to share with specific people will find its way to other people
rsk2 People will share information about me without my consent, for example by tagging me in pictures
rsk3 I will often be exposed to information I don't care about
rsk4 Commercial companies will use information I share to my disadvantage, for example by bombarding me with irrelevant advertisements
rsk5 Commercial companies will sell my information to third parties without my knowledge or consent
rsk6 The state will know more about me than I want it to
rsk7 Hackers will steal my information

Significance

	Min.	Max.	Mean	Std. D.	Skewness	Std. Error	Kurtosis	Std. Error
rsk1	1	5	3.61	1.35	−.62	.09	−.81	.18
rsk2	1	5	3.45	1.36	−.36	.09	−1.12	.18
rsk3	1	5	3.27	1.28	−.26	.09	−.94	.18
rsk4	1	5	3.94	1.20	−.96	.09	−.04	.18
rsk5	1	5	4.02	1.26	−1.10	.09	.02	.18
rsk6	1	5	3.43	1.37	−.39	.09	−1.05	.18
rsk7	1	5	4.04	1.32	−1.11	.09	−.09	.18

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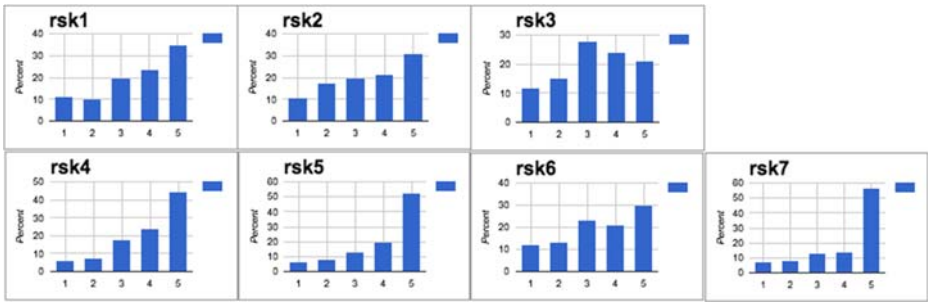
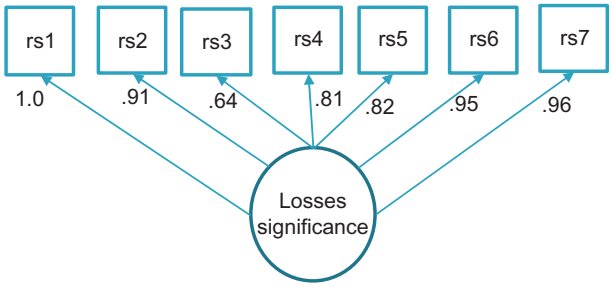


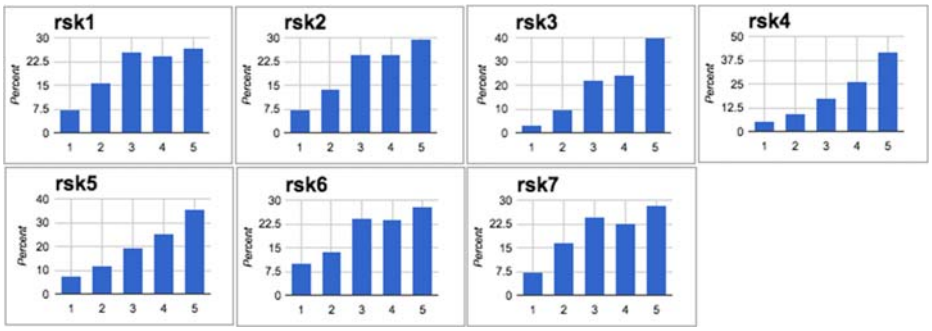
Fit of model:

$\chi^2 = 8.0$, $df = 6$, $p = .24$; RMSEA = .02, CFI = .997



Likelihood

	N	Min.	Max.	Mean	Std. D.	Skewness	Std. Error	Kurtosis	Std. Error
rsk1	743	1	5	3.47	1.24	−.34	.09	−.91	.18
rsk2	743	1	5	3.56	1.25	−.46	.09	−.81	.18
rsk3	743	1	5	3.87	1.15	−.71	.09	−.47	.18
rsk4	743	1	5	3.9	1.19	−.87	.09	−.23	.18
rsk5	743	1	5	3.71	1.27	−.66	.09	−.66	.18
rsk6	743	1	5	3.46	1.30	−.41	.09	−.92	.18
rsk7	743	1	5	3.49	1.26	−.33	.09	−.98	.18



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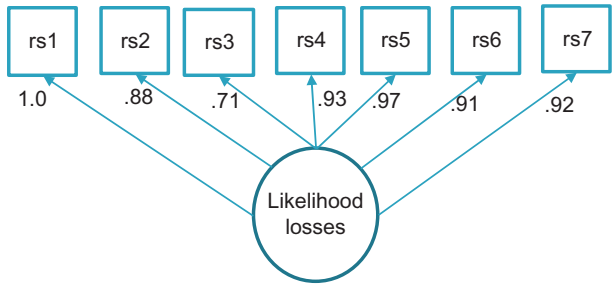
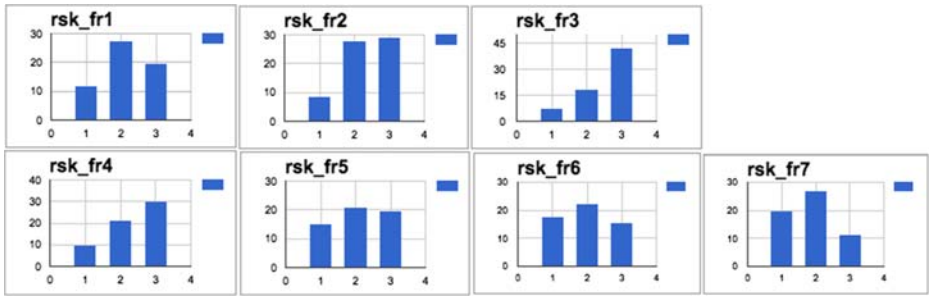
$\chi^2 = 3.8$, $df = 6$, $p = .73$; RMSEA = .00, CFI = .999



Losses—others

rsk_fr1 Information they want to share with specific people will find its way to other people
rsk_fr2 People will share information about them without their consent, for example by tagging them in pictures
rsk_fr3 They will often be exposed to information they don't care about
rsk_fr4 Commercial companies will use information they share to their disadvantage, for example by bombarding them with irrelevant advertisements
rsk_fr5 Commercial companies will sell their information to third parties without their knowledge or consent
rsk_fr6 The state will know more about them than they want it to
rsk_fr7 Hackers will steal their information

	N	Min.	Max.	Mean	Std. D.	Skewness	Std. Error	Kurtosis	Std. Error
rsk_fr1	440	1	3	2.13	.72	−.20	.12	−1.06	.23
rsk_fr2	490	1	3	2.31	.69	−.50	.11	−.83	.22
rsk_fr3	509	1	3	2.51	.68	−1.07	.11	−.12	.22
rsk_fr4	454	1	3	2.34	.74	−.63	.12	−.92	.23
rsk_fr5	418	1	3	2.08	.79	−.14	.12	−1.37	.24
rsk_fr6	413	1	3	1.96	.77	.06	.12	−1.32	.24
rsk_fr7	433	1	3	1.85	.72	.23	.12	−1.04	.23



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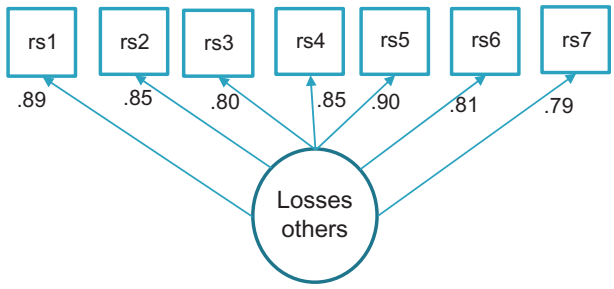
$\chi^2 = 13.0$, $df = 9$, $p = .15$; RMSEA = .02, CFI = .992



Construct validity

To validate our measurements, we tested their correlations with related concepts on an additional sample, relying on previous research by Dienlin and Metzger (2016), who developed scales to measure Facebook benefits, privacy concerns, Facebook privacy self-efficacy, Facebook self-disclosure and Facebook self-withdrawal. Furthermore, to validate data about self-disclosure, we asked participants to provide us with their Facebook activity log and added an item measuring passive Facebook use (demonstrating distinction between self-disclosure and passive use).

Sample

We used an online sample of 261 participants, 124 of whom completed the survey, ranging in age between 19 and 64 with $M = 40.5$ $SD = 12.7$; 49% male; 3% ultra-Orthodox; 50% with academic degrees; income: $M = 3.0$ $SD = 1.4$ on a 5-point scale; participants were recruited using the same platform as the samples described in the article.

Measures

In addition, developing the above measures, we also asked participants about their Facebook benefits, privacy concerns, Facebook privacy self-efficacy, Facebook self-disclosure and Facebook self-withdrawal. All scales are taken from Dienlin and Metzger (2016) (full scales: https://osf.io/h4pef/?view_only=cf4c10222def4efdbecae20c1dca7dc6).

Activity log data: Facebook allows users to download a log documenting all their activities. To protect participants' privacy, we did not ask them to disclose their log (containing a complete documentation of photos, posts, etc.), but rather to report the size of the files downloaded (see Figure A1), based on the assumption that size is an indicator of amount of material shared. Nevertheless, it is important to note that 137 participants did not comply with this request, resulting in a high attrition rate. Furthermore, 22 participants failed to follow the instructions and thus supplied incorrect data. Seeking a uniform size indicator for all files (photos, posts, events, friends, etc.), we standardized each item (image files are much larger than text files, for example), ensuring that each participant was assigned an accurate relative score that could be compared with those of other participants.

Passive use: To gauge passive use, we asked participants how often they read their Facebook friends' posts (text and images) as part of the activity scale we developed to measure SD act frequency.

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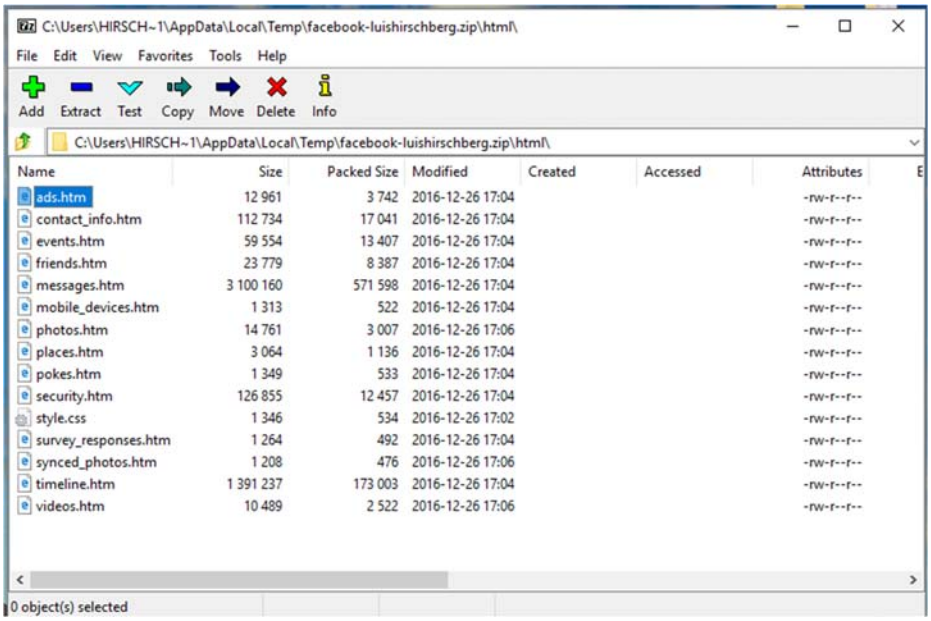

Figure A1 Sample activity log data disclosed by participant.

Results

Table A1 presents the descriptive statistics of variables measured, while Table A2 displays the results of a factor analysis for all scales. As indicated, all items in our scales were loaded on one factor, except for the SD acts scale, for which a two-factor solution was identified: SD acts were loaded on one factor and another measuring passive use was loaded on another. Only one item displayed high loadings of both factors: Liking friends' posts/images. The dual high score may be explained by the item's relevance to self and others, as it measures the participant's active use yet relies on observation by others as well.

Table A3, that presents the correlation among the variables, provides additional validation for the measurements developed: Self-disclosure measurements were highly correlated with Dienlin and Matzear's Facebook self-disclosure measurements, suggesting convergence validity. We also noted a positive correlation among perceived gains, Dienlin and Matzear's Facebook benefits, self-disclosure and privacy concerns. Perceived losses were correlated with Facebook self-withdrawal and privacy concerns.

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Table A1 Descriptive Statistics of All Variables

	Mean	Std. D.	Skewness	S.E.	Kurtosis	S.E.	Cronbach's α
SD activity	2.43	.85	.66	.21	.11	.43	.89
SD information	1.78	.74	1.54	.23	3.10	.46	.88
Gains	3.50	1.00	−.34	.23	−.40	.46	.86
Risks	3.60	.95	−.48	.23	−.25	.46	.86
Privacy concerns (Dienlin and Matzear)	2.75	.79	−.35	.23	−.32	.46	.59
Facebook privacy self-efficacy (Dienlin & Matzear)	2.48	.96	.25	.23	−.63	.46	.88
Facebook benefits (Dienlin and Matzear)	3.29	.86	−.37	.23	.63	.46	.90
Facebook self-disclosure (Dienlin and Matzear)	2.68	.94	.24	.23	−.26	.46	.86
Facebook self-withdrawal (Dienlin and Matzear)	1.69	.29	−1.07	.23	.54	.46	.61
Activity log data	.002	.66	1.04	.24	2.02	.47	.87

Type	N	Min.	Max.	Mean	Std. D.
Ads	101	2	62,919	9,547	10,773
Contact info	102	1.61	2,879,918	145,797	361,512
Events	102	2	20,187,833	329,373	2,032,835
Friends	102	2	191,275	13,994	28,121
Messages	101	1.01	43,798,254	3,173,460	6,615,034
Mobile device	102	1	1,538	858	681
Photos	102	1	65,703	6,687	11,083
Places	102	1	10,628	1,082	1,517
Synced photos	102	1	1,431	792	628
Timeline	102	3	8,214,581	569,150	1,050,921
Videos	102	1	42,768	4,305	7,629

Table A2 Exploratory Factor Analysis

SD actions Item	Component	
	1	2
Write status updates	.78	−.02
Comment on other people's status updates	.73	.38
Share other people's content	.73	.32
Upload photos	.79	−.16
Tag yourself in photos	.79	−.39

(*Continued*)

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Table A2 Continued

SD actions Item	Component	
Tag other people in photos	.78	−.30
Participate in groups	.63	.45
Indicate your geographical location, for example by mentioning a name of a restaurant, an event, or a landmark	.76	−.36
Indicate you intend to go to an event, or check in	.74	−.16
Read content uploaded by your friends	.44	.62

SD information Item	Component
Share their opinions regarding current events and political contents they support	.835
Share content regarding their consumer preferences and shopping habits, such as companies and products they like or dislike	.814
Share information regarding their family, such as their relationship status, the identity of their partner, their parental status, or activities in which their children took part	.778
Share information regarding their work, such as their workplace, their profession, or the people they work with	.747
Share information regarding their friends, such as the identity of the people they hang out with or how they hang out with them	.743
Share information regarding their current location, such as restaurants or events they are currently attending	.705
Share information regarding places they previously attended, such as restaurants and events	.675
Indicate where they live on their profile	.62
Share personal information, such as phone numbers, ID numbers, or passwords	.562

SD gains	
Significance	
Item	Component
Simply and rapidly obtaining information suiting my interests	.687
Obtaining likes and support from others	.845
Expressing myself—my opinions and feelings	.818
Maintaining and reinforcing social relations	.696
Getting personalized information, such as information about my close friends or about my fields of interest	.794
Effectively promoting myself professionally	.756

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Likelihood	
Item	Component
Simply and rapidly obtaining information suiting my interests	.816
Obtaining likes and support from others	.865
Expressing myself—my opinions and feelings	.779
Maintaining and reinforcing social relations	.733
Getting personalized information, such as information about my close friends or about my fields of interest	.719
Effectively promoting myself professionally	.631

SD losses	
Significance	
Item	Component
Information I want to share with specific people will find its way to other people	.82
People will share information about me without my consent, for example by tagging me in pictures	.714
I will often be exposed to information I don't care about	.523
Commercial companies will use information I share to my disadvantage, for example by bombarding me with irrelevant advertisements	.701
Commercial companies will sell my information to third parties without my knowledge or consent	.823
The state will know more about me than I want it to	.779
Hackers will steal my information	.781

Likelihood	
Item	Component
Information I want to share with specific people will find its way to other people	.865
People will share information about me without my consent, for example by tagging me in pictures	.815
I will often be exposed to information I don't care about	.734
Commercial companies will use information I share to my disadvantage, for example by bombarding me with irrelevant advertisements	.82
Commercial companies will sell my information to third parties without my knowledge or consent	.816
The state will know more about me than I want it to	.717
Hackers will steal my information	.775

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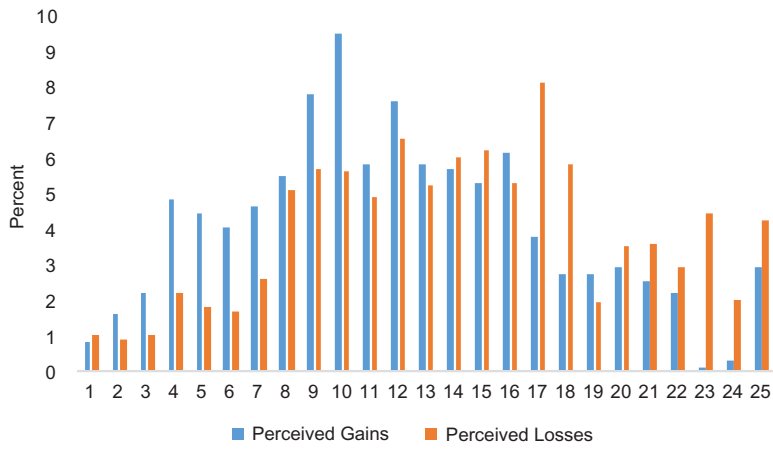
Table A3 Pearson's Correlations for Measurements

	Acts	Information	Gains-sig	Gains-lik	Loss-sig	Loss-sig	Activity log	Self-withdrawal	Benefits	SD
Information	.754**									
	121									
Gains-sig	.487**	.411**								
	121	121	121							
Gains-lik	.591**	.516**	.758**							
	121	121	121							
Loss-sig	.236**	.239**	.393**	.433**						
	121	121	121	121						
Loss-lik	.301**	.194*	.419**	.407**	.586**					
	121	121	121	121	121					
Activity log	**.274****	**.212***	.349**	.347**	.219*	.321**				
	102	102	102	102	102	102				
Self-withdrawal	.152	.159	.108	.165	**.235***	**.221***	.168			
Dienlin and Matzear	119	119	119	119	119	119	100			
Benefits	.168	.212*	**.342****	**.404****	−.011	−.007	−.022	.161		
Dienlin and Matzear	121	121	121	121	121	121	102	119		
SD	**.556****	**.573****	.531**	.622**	.285**	.210*	.210*	.145	.349**	
Dienlin and Matzear	121	121	121	121	121	121	102	119	121	
Privacy con	−.044	−.098	−.06	.008	**.148**	**.341****	.216*	.198*	−.275**	−.118
Dienlin and Matzear	121	121	121	121	121	121	102	119	121	121

*p < .5; **p < .01

Appendix B: Distribution of Risk Assessment Measures



Note: Percent refers to the percentage of participants for each scale. Perceived gains are measured as the likelihood of gain (on a scale of 1–5) weighted by its magnitude (on a 1–5 scale). Perceived losses measured as the likelihood of loss (on a scale of 1–5) weighted by its magnitude (on a scale of 1–5).

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Appendix C: Pearson's Correlations Among Variables

	1	2	3	4	5	6	7	8	9	10	11
2. Sharing information—self	.665**										
	744										
3. Sharing activity—others	.278**	.197**									
	774	724									
4. Sharing information—others	.126**	.321**	.592**								
	744	704	744								
5. Browsing	.231**	.126**	.058	.093*							
	732	742	732	704							
6. Facebook use (hrs.)	.473**	.335**	.103**	.126**	.425**						
	730	742	730	704	742						
7.Other SNS use (hrs.)	.344**	.359**	.254**	.132**	.218**	.339**					
	240	742	240	704	742	742					
8. Past damage—self	.115**	.222**	.222**	.119**	.086*	.06	.166**				
	742	742	742	704	742	742	742				
9. Losses—others	−.004	.166**	.109**	.130**	.015	.048	.154**	.447**			
	622	742	619	704	742	742	742	742			
10. Losses—self	.038	−.04	.063	.032	.009	−.029	.022	.281**	.046		
	742	742	724	704	742	742	742	742	742		
11. Gains—self	.538**	.391**	.272**	.158**	.082*	.369**	.198**	.036	.059	.076*	
	741	742	723	704	742	742	742	742	742	742	
12. Perceived gains—others	.201**	.132**	.400**	.258**	.046	.154**	−.037	−.063	-.083*	.095*	.507**
	616	632	616	623	632	632	632	632	632	632	632

Notes: *p < .5; **p < .01

Variables: (1) Sharing activity—self; (2) Sharing information—self; (3) Sharing activity—others; (4) Sharing information—others; (5) Browsing; (6) Facebook use in hours; (7) Other SNS use in hours; (8) Sustained damage in the past—self; (9) Losses—others; (10) Losses—self; (11) Gains –self; (12) Perceived gains—others.

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Appendix D: Manipulation Check

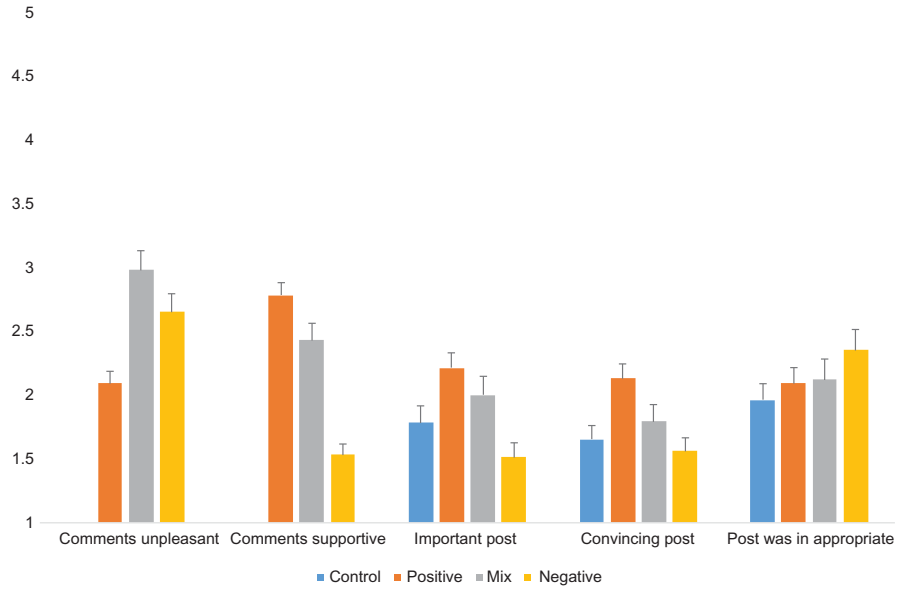

Figure A Means and std. errors for post and comment evaluation, according to experimental condition: 1 (disagree); 5 (agree).

Appendix E

Table 1 Results of Contrast between Experimental Conditions

	Contrast	Value of Contrast	Std. Error	t	Sig.
Self-disclosure	Control vs. Negative	.34	.13	2.60	.01
	Control vs. Positive	−.31	.12	−2.57	.01
	Mix vs. Negative	.33	.13	2.48	.01
	Mix vs. Positive	−.32	.12	−2.55	.01
	Negative vs. Positive	−.32	.12	−2.55	.01
Gains	Control vs. Negative	.42	.17	2.42	.02
	Control vs. Positive	−.24	.16	−1.48	.14
	Mix vs. Negative	.25	.18	1.38	.17
	Mix vs. Positive	−.41	.17	−2.49	.01
	Negative vs. Positive	−.41	.17	−2.49	.01

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